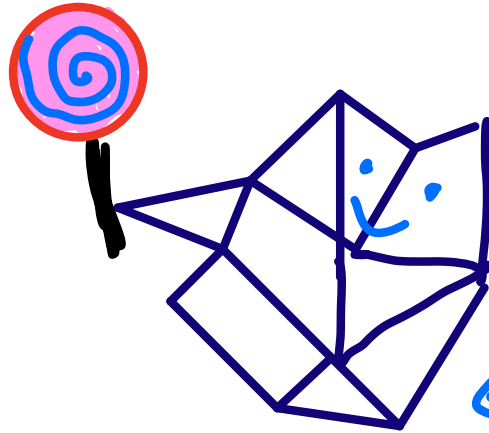


# Cosmic TENSION & Planck Scale. Messages



↑ Angry  
mis-understood  
Cosmos



a model of  
quantum geometry  
Showing Care

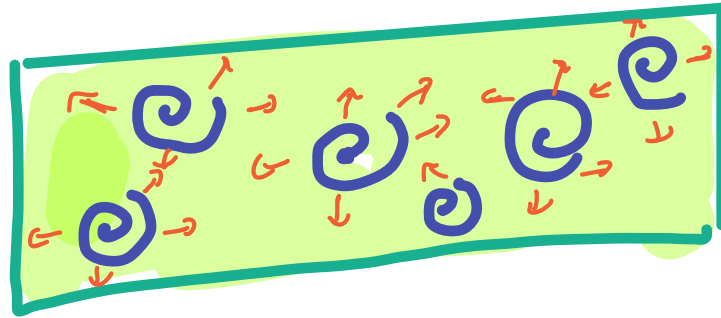
John Joseph Marchetta

"Alleviating the Tension in the Cosmic Microwave  
Background using Planck-Scale Physics" - Ashtekar  
2020

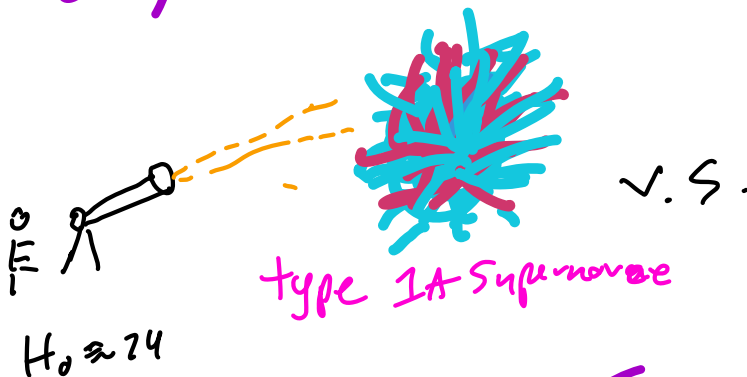
" $T_0$   $H_0$  or not to  $H_0$ " - George Efsthion 2021

# Outline

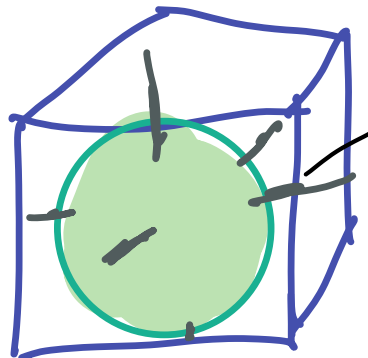
## 1.) Expanding Space



## 2.) Hubble tension

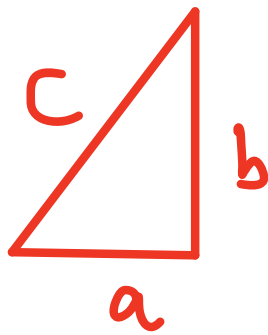


## 3.) Planck Era Predictions



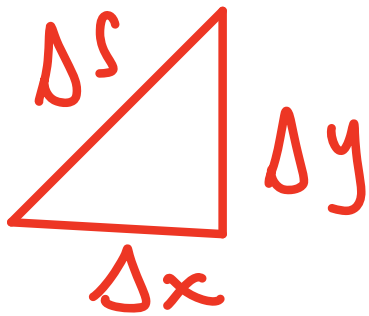
$$6 l_p^2 = \text{Area}$$

# What's a Metric?



$$a^2 + b^2 = c^2$$

Physics Notation:

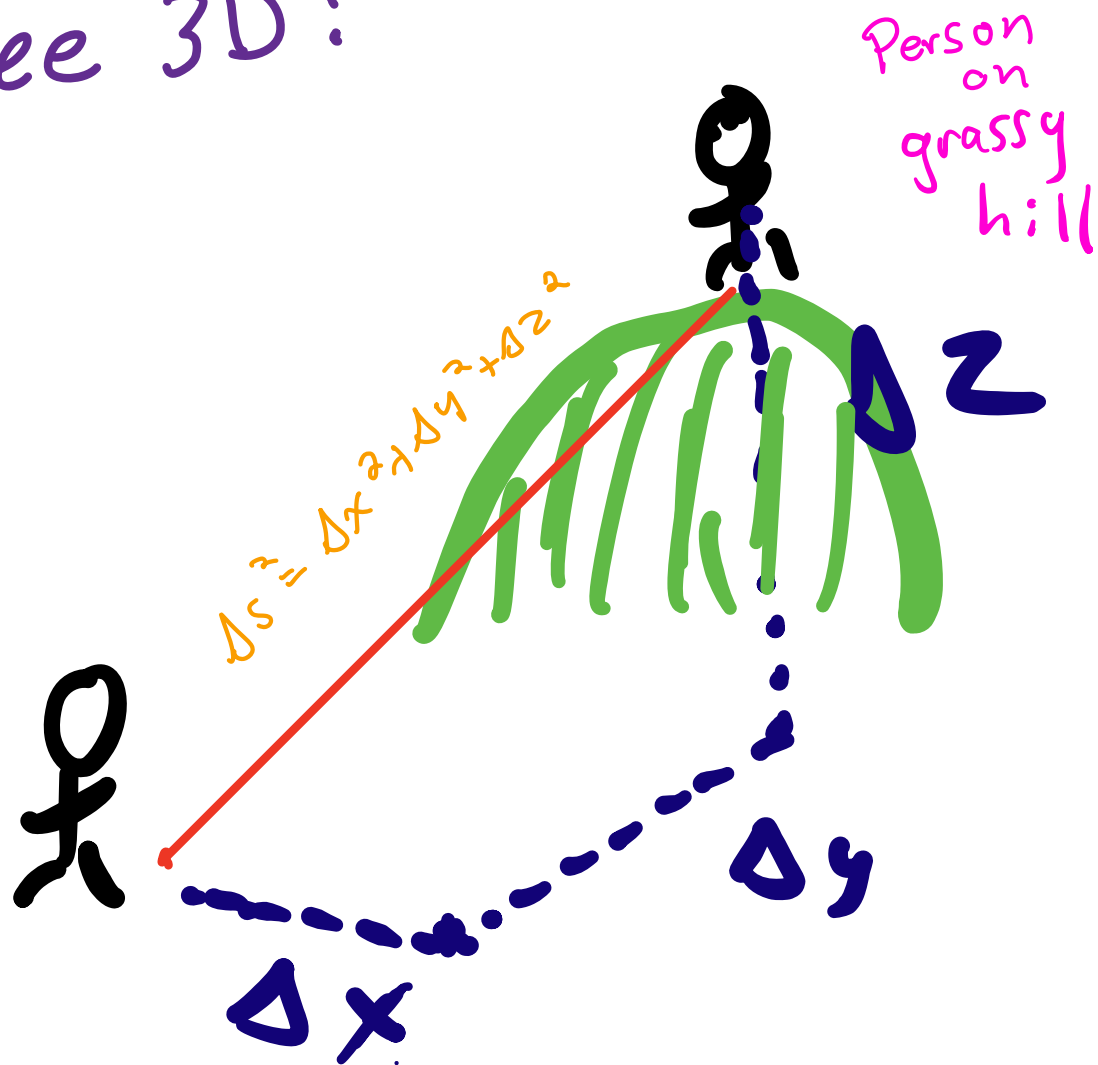


$$(\Delta s)^2 = (\Delta x)^2 + (\Delta y)^2$$

Infinitely small triangles

$$ds^2 = dx^2 + dy^2$$

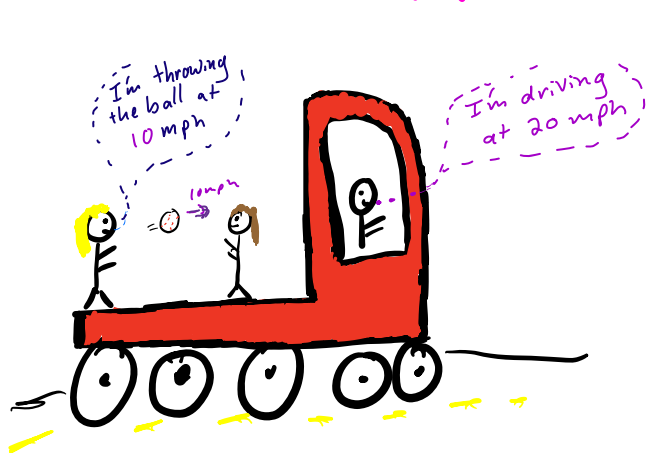
Looking around we  
see 3D:



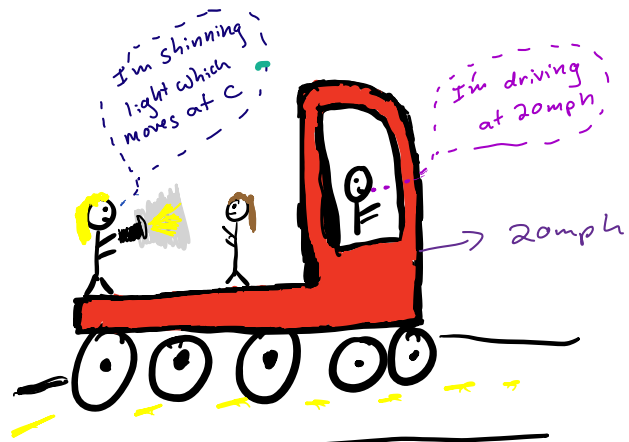
$$(ds)^2 = (\Delta x)^2 + (\Delta y)^2 + (\Delta z)^2$$

$$ds^2 = dx^2 + dy^2 + dz^2$$

Since the speed of light being  
Constant is not compatible with  
Galilean relativity



"I'm sitting in the grass!"  
 $10 \text{ mph} + 20 \text{ mph} = 30 \text{ mph}$



"I'm sitting in the grass!"  
 $c + 20 \text{ mph} = c?$

Einstein said we live in a 4D universe

3D:  $ds^2 = dx^2 + dy^2 + dz^2$

4D:  $ds^2 = dx^2 + dy^2 + dz^2 - c^2 dt^2$

We live in an expanding Universe:

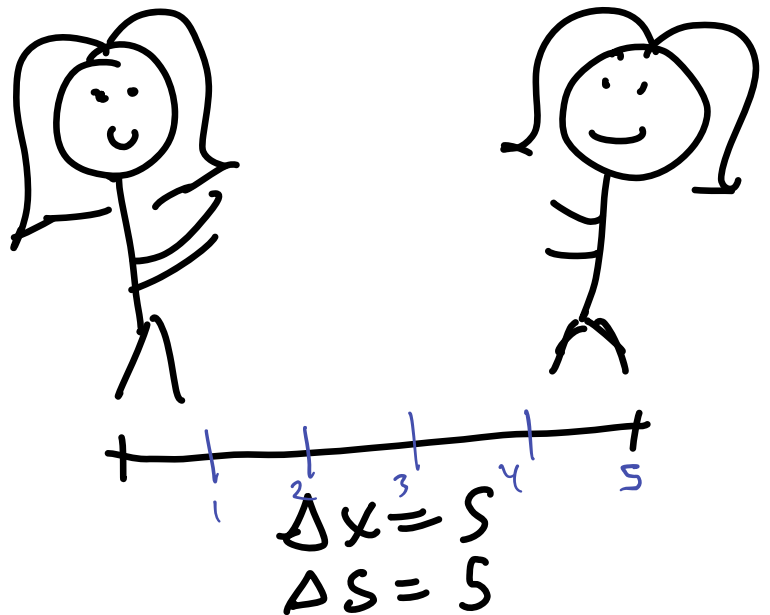
$$(\Delta S)^2 = a(t) \left( (\Delta x)^2 + (\Delta y)^2 + (\Delta z)^2 \right) - c^2 \Delta t^2$$

take:  $a(t) = t$

at  $t=1$   
 $\Delta x = 5$

$$(\Delta S)^2 = (\Delta x)^2$$

$$\Rightarrow \Delta S = 5$$



# Evolving forward in time

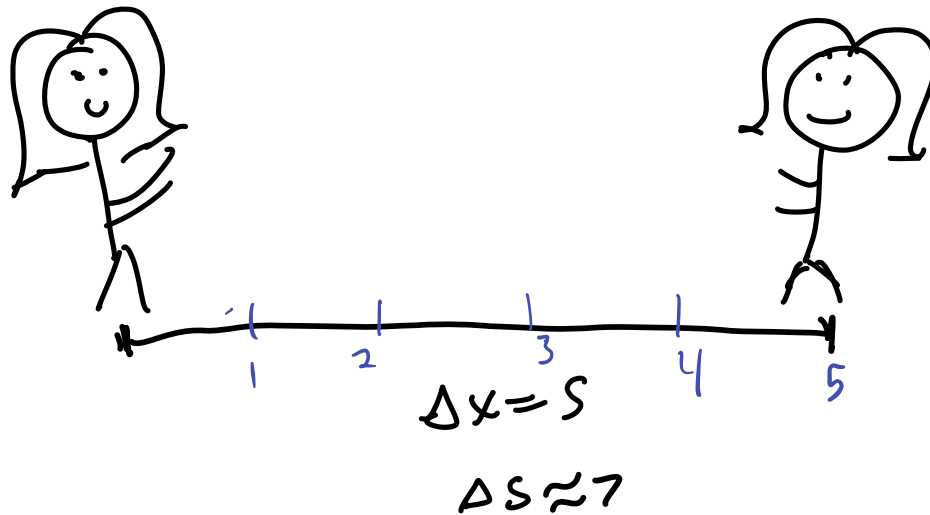
$$(\Delta s)^2 = a(t) \left( (\Delta x)^2 + (\Delta y)^2 + (\Delta z)^2 \right) - c^2 \Delta t^2$$

at  $t=2$

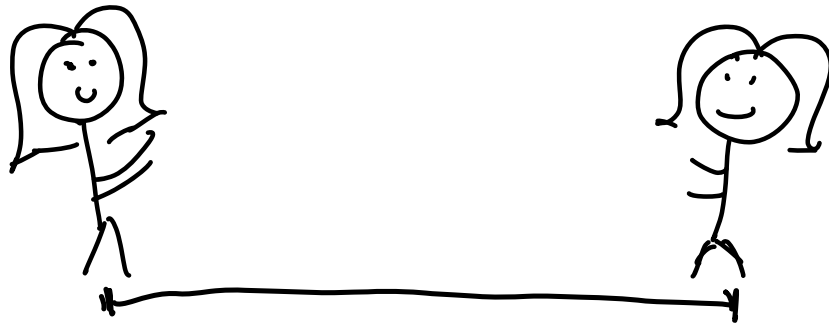
Still  $\Delta x = 5$

$$(\Delta s)^2 = 2 (\Delta x)^2$$

$$(\Delta s) = \sqrt{50} \approx 7$$



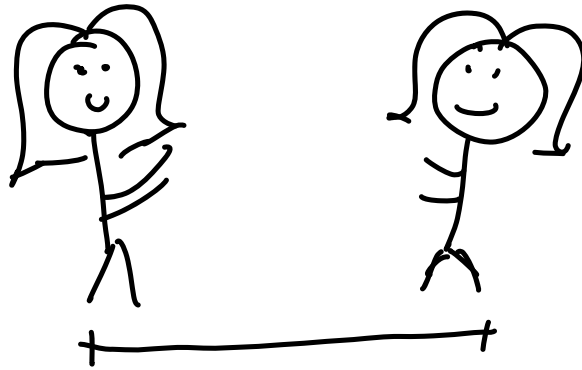
# Evolving backward in time



$$T = 2$$

$$\Delta x = 5$$

$$\Delta S \approx 7$$



$$T = 1$$

$$\Delta x = 5$$

$$\Delta S = 5$$

Big  
Bang

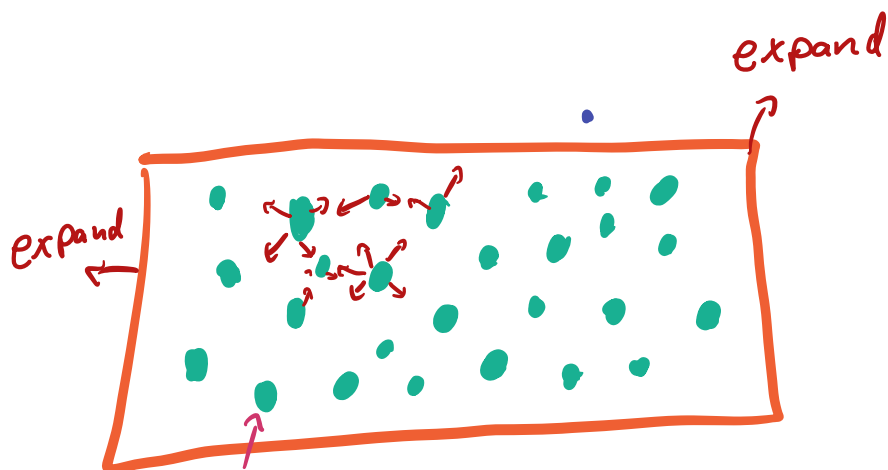


$$\Delta x = 5$$

$$\Delta S = 0$$

$$T = 0$$

# Cosmic Expansion



Like a box of gas

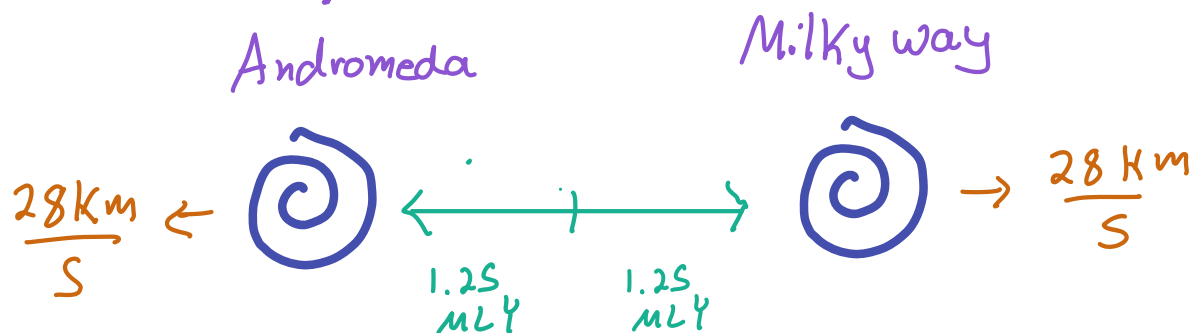
1 of about  
2,000,000,000,000 galaxies

Define:  $V = H d$   
 speed  $\downarrow$   $\uparrow$  distance  
 $\uparrow$   
 hubble constant

$$H = \frac{\dot{a}}{a} = \frac{V}{d}$$

$$\dot{a} = \frac{da}{dt}$$

$$H \approx 73.4 \frac{\text{km}}{\text{s}} \text{MPC}$$



$t < 10^{-36}$   
seconds

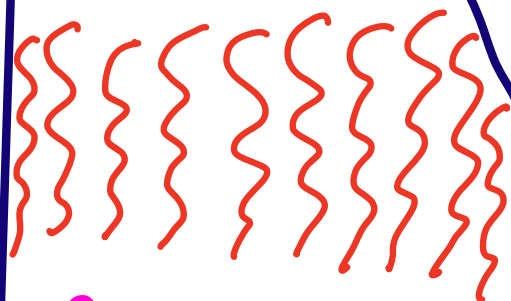
Planck era  
Grand Unification

$10^{-36}$   
seconds

Inflation

$< t <$   
 $10^{-32}$   
seconds

$$\frac{n_p}{v} = \frac{n_{\bar{p}}}{v} \approx 0$$



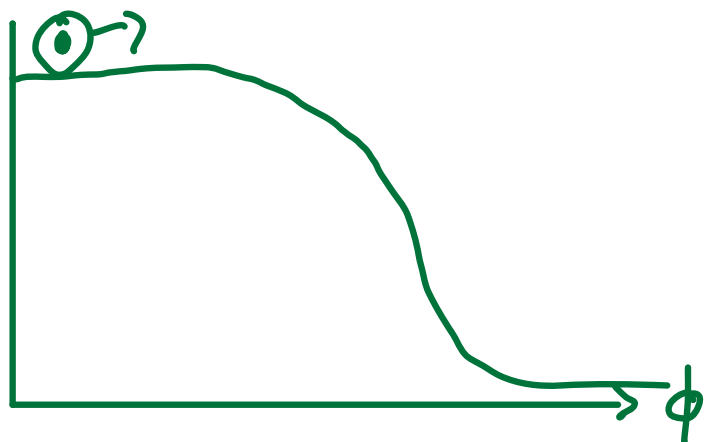
$$\frac{n_p}{v} = \frac{n_{\bar{p}}}{v} \neq 0$$

$t < 10^{-14}$   
seconds

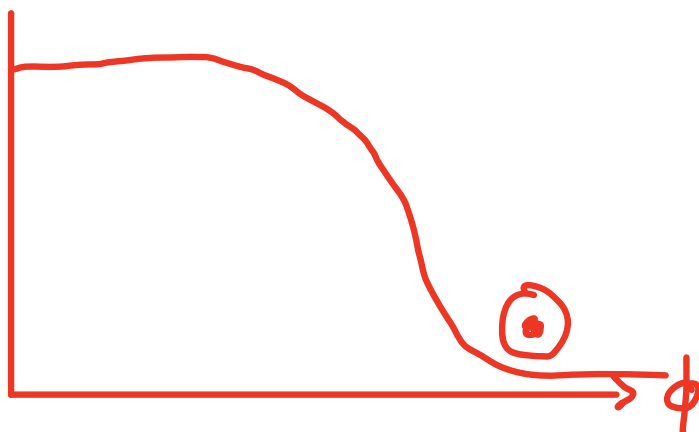
$$\frac{n_p}{v} > \frac{n_{\bar{p}}}{v}$$

Electroweak  
Baryogenesis

$v(\phi)$



$v(\phi)$



E.O.M

$$\ddot{\phi} + H \dot{\phi} + V'(\phi) = 0$$

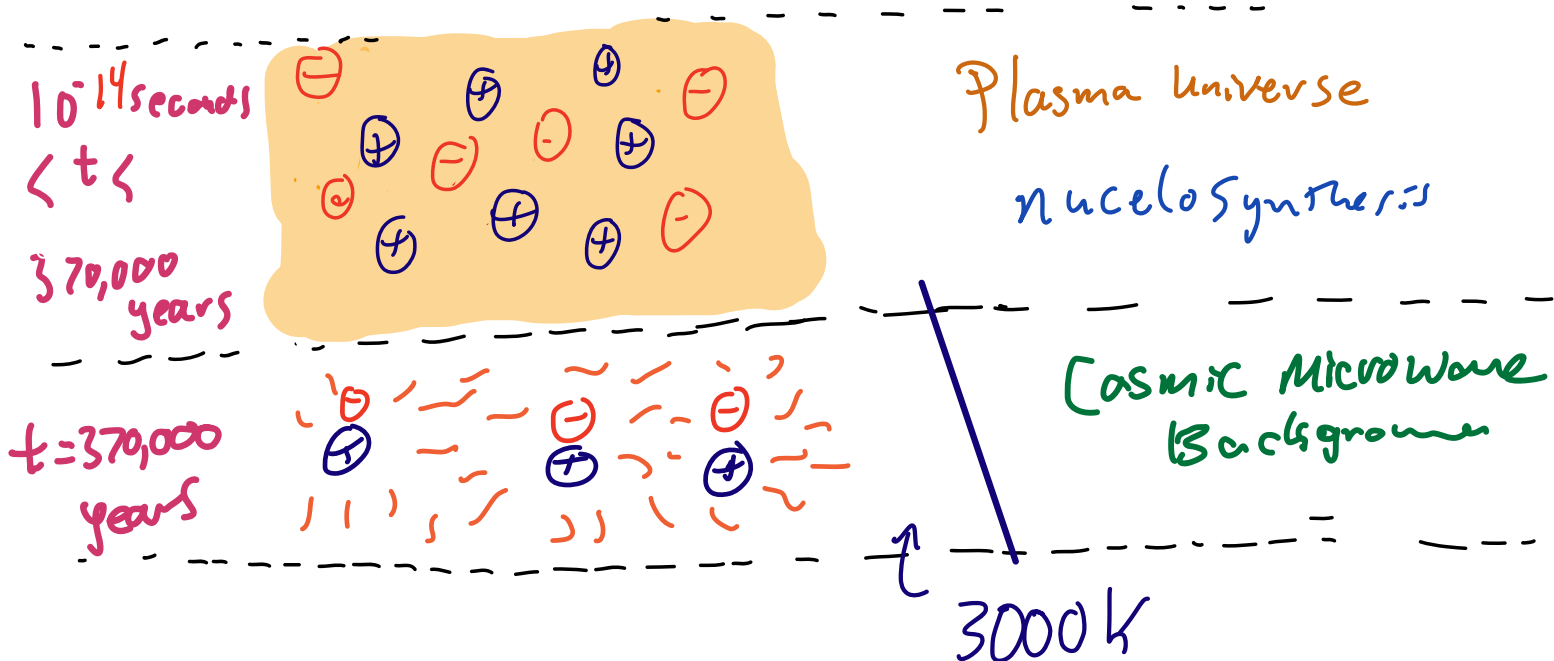
$$\ddot{x} + \gamma \dot{x} + Kx = 0$$

Strongly Damped

$$|V(\phi)''| \ll H$$

$$\omega = \frac{\dot{\phi} - V}{\dot{\phi} + V} \approx -1$$

$$P = \omega p$$



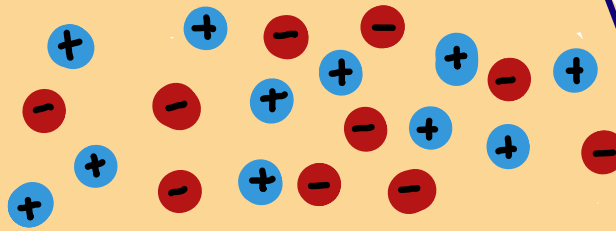


$t < 10^{-14}$   
seconds

$$\frac{n_p}{v} > \frac{n_{\bar{p}}}{v}$$

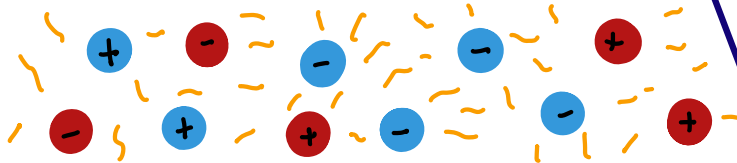
Electroweak  
Baryogenesis

$10^{-14}$  seconds  $<$   
 $t < 370,000$   
years



Nucleosynthesis  
Plasma

$t = 370,000$   
years



CMB

# History of the Universe

$t < 10^{-36}$   
Seconds

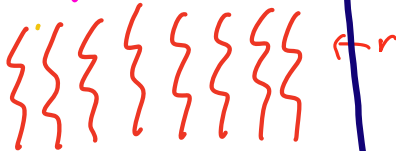


Planck era  
Grand Unification era

$t < 10^{-43}$   
 $t < 10^{-35}$

$10^{-36}$   
second  
 $t <$

density:  
 $\frac{n_p}{V} = \frac{n_{\bar{p}}}{V} \approx 0$



$\frac{n_p}{V} = \frac{n_{\bar{p}}}{V} \neq 0$

$10^{-32}$   
Second

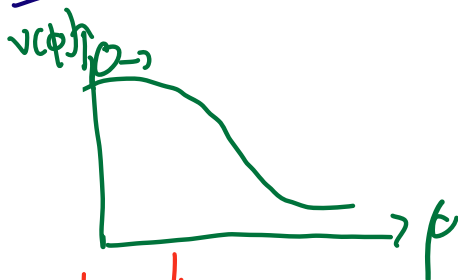
$t < 10^{-14}$   
Seconds

$\frac{n_p}{N} > \frac{n_{\bar{p}}}{V}$

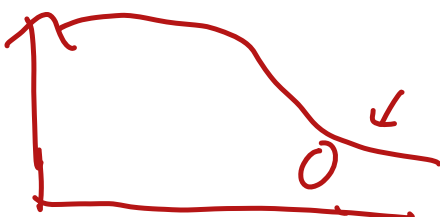
Electroweak baryogenesis  
 $B-L \rightarrow B$   $B-L$  symmetry

E.O.M

Inflation



heating  
 $u(\phi)$



$$\ddot{\phi} + H\dot{\phi} + V'(\phi) = 0$$

Harmonic oscillator

$$\ddot{x} + \gamma\dot{x} + kx = 0$$

Strongly damped

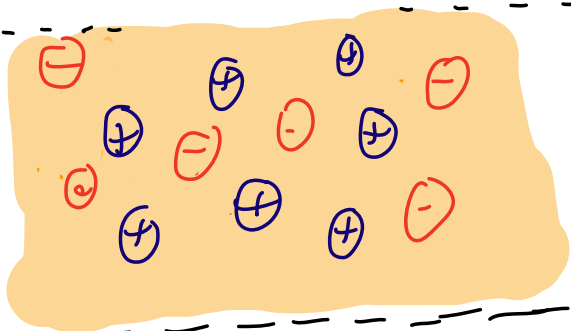
$$|V''(\phi)| \ll H$$

$$w = \frac{\dot{\phi}^2 - V}{\dot{\phi}^2 + V} \approx -1$$

$10^{-14}$  seconds  
 $< t <$

370,000  
years

$t = 370,000$   
years



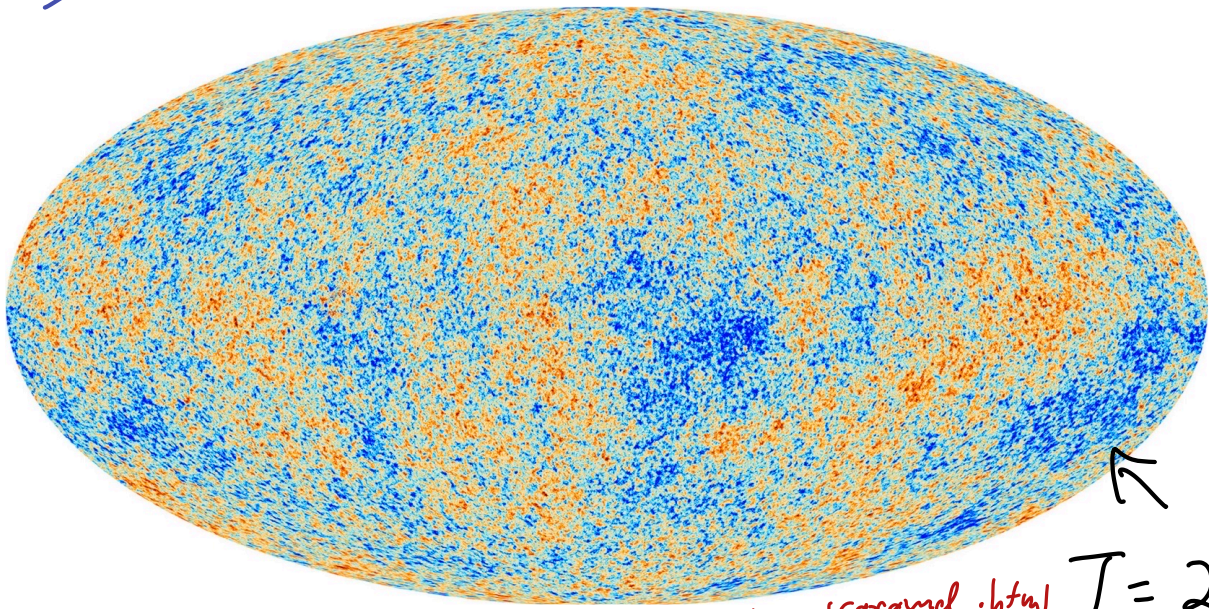
Plasma universe

nucleosynthesis

Cosmic Microwave  
Background

3000 K

# Cosmic Microwave Background



Space.com / 33982-cosmic-microwave-background.html  $T = 2.73 \text{ K}$

Anisotropies  $\frac{\Delta T}{T} = 10^{-3} \text{ K}$

Where do these anisotropies come from?

$\frac{k}{a} < H$  (wavelength is larger than hubble radius)

$\psi_k = c \frac{H}{k} e^{i\vec{k} \cdot \vec{x}}$

$\frac{k}{a} > H$  (wavelength is smaller than hubble radius)

$\psi_k = c \frac{1}{a(t)} e^{i k a t} e^{i\vec{k} \cdot \vec{x}}$

# Perturbations of Inflation

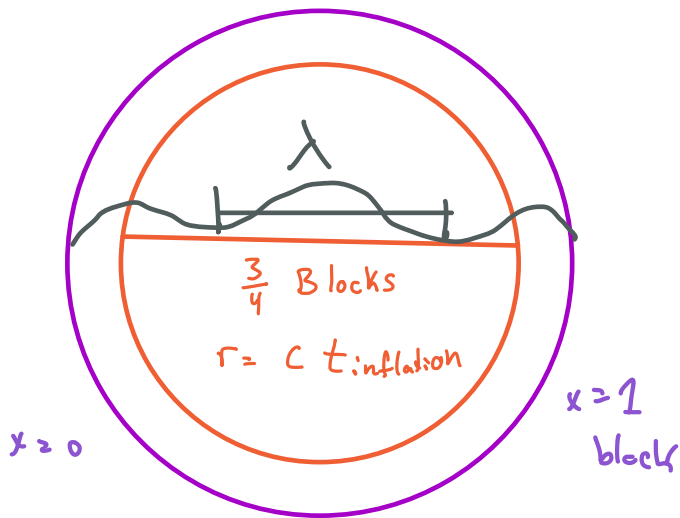
$\frac{k}{a} < H$  (Wavelength longer than Hubble radius)

$$\psi_k = c \frac{H}{k} e^{i\vec{k} \cdot \vec{x}}$$

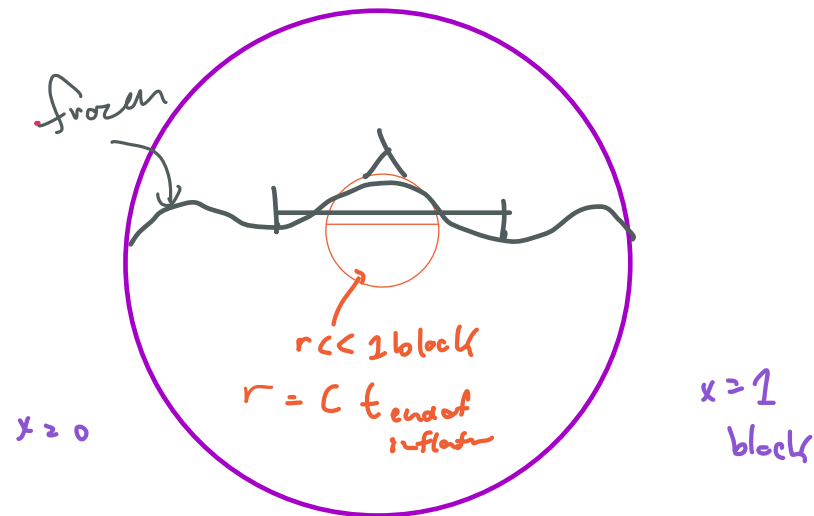
$\frac{k}{a} > H$  (Wavelength shorter than Hubble radius)

$$\psi_k = \frac{c}{a(t)} e^{i k a t} e^{i\vec{k} \cdot \vec{x}}$$

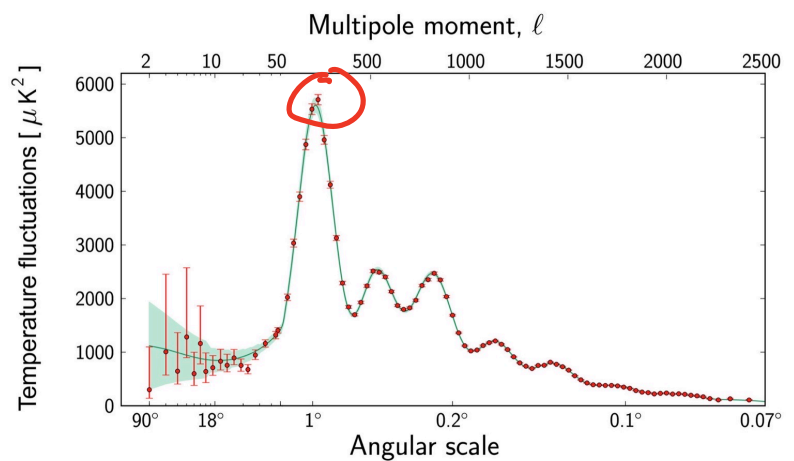
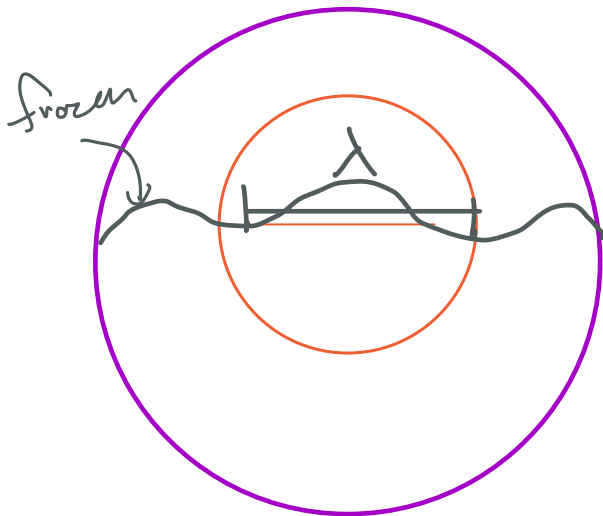
Beginning of inflation



End of inflation



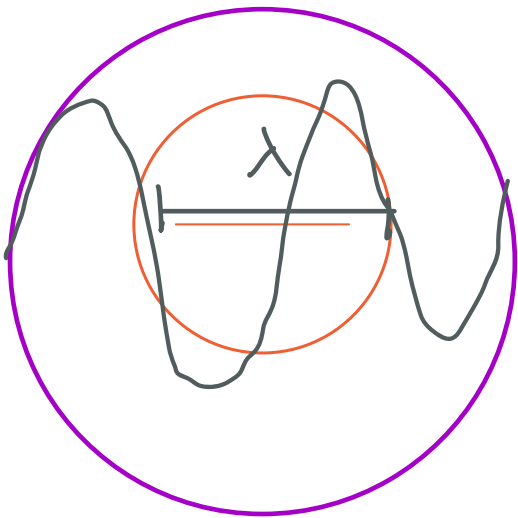
$t = 370,000$  years



$\frac{k}{a} < H$  (Wavelength longer than Hubble radius)

$$\psi_k = c \frac{H}{k} e^{i\vec{k} \cdot \vec{x}}$$

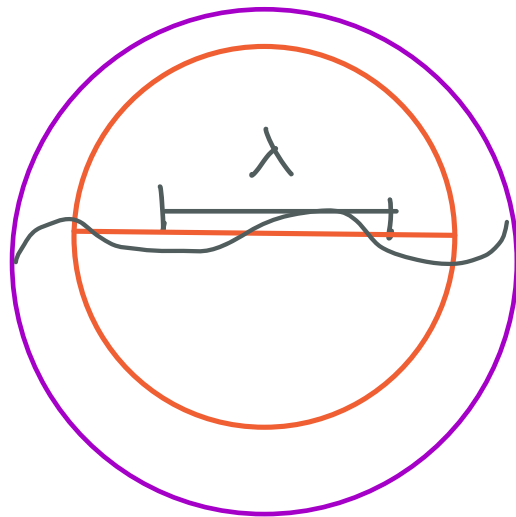
$t_{\text{end of inflation}} < t_1 < t_{\text{CMB}}$



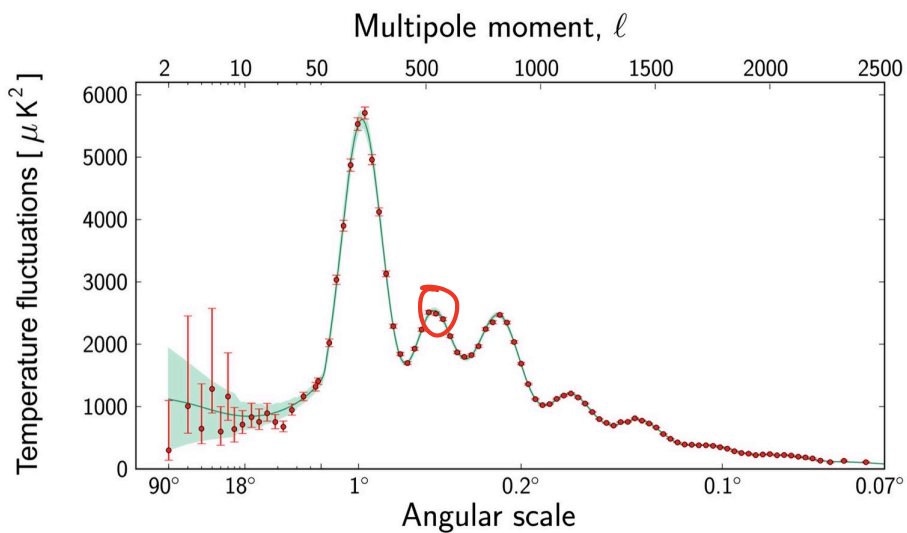
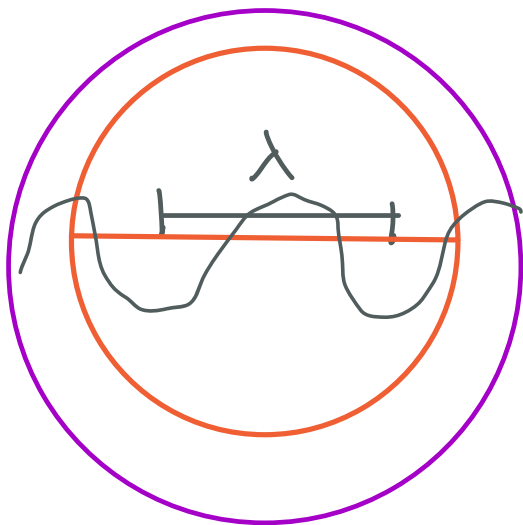
$\frac{k}{a} > H$  (Wavelength shorter than Hubble radius)

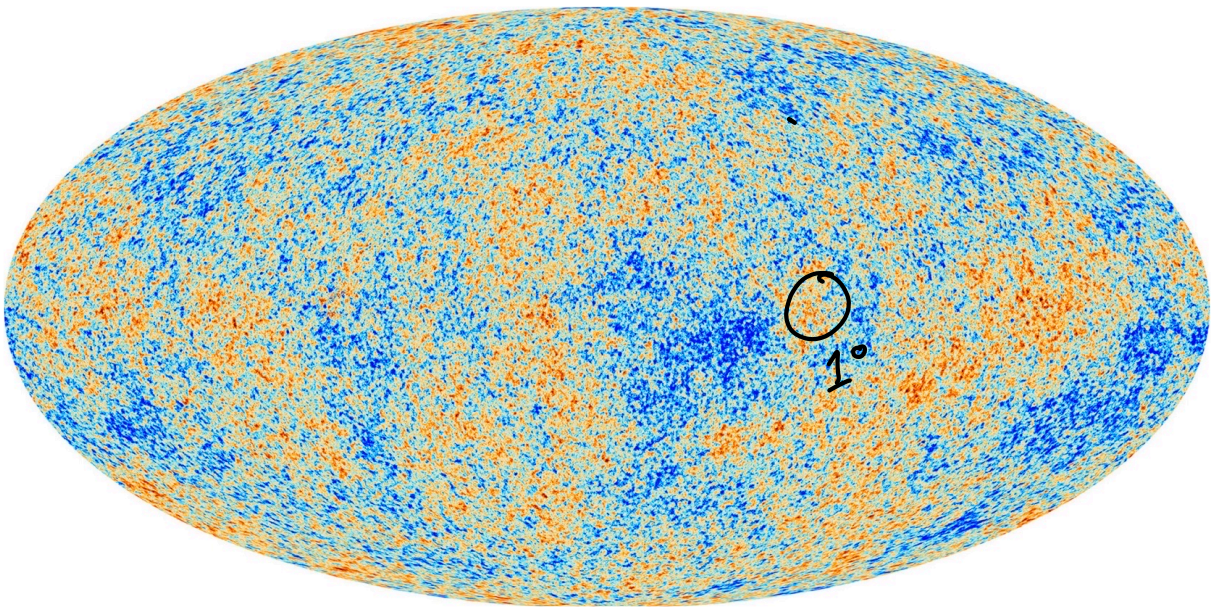
$$\psi_k = \frac{c}{a(t)} e^{i k a t} e^{i \vec{k} \cdot \vec{x}}$$

$t_1 < t < t_{\text{CMB}}$



$t = 370,000 \text{ year}$





Infer  $H(z)$  from size of CMB Anisotropies

$$H(z) = H_0^f \left[ \left( \Omega_m (1+z)^3 + (1-\Omega_m) \left[ 1 + \Delta \exp(- (z/z_c)^\beta) \right] \right) \right]^{1/2}$$

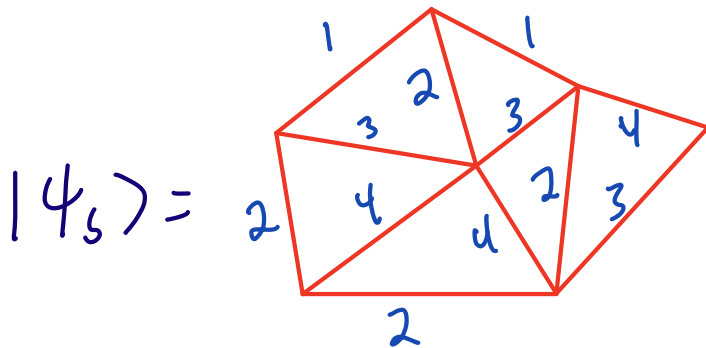
Early time :  $H_0 = 67.44 \pm 0.58 \frac{\text{km}}{\text{s MPC}}$   
CMB

Late time :  $H_0 = 74.03 \pm 1.42 \frac{\text{km}}{\text{s MPC}}$   
Cepheids

# Loop Quantum Gravity

QM:  $H|\psi\rangle = (n + \frac{1}{2})\hbar\omega|\psi\rangle$

LQG:  $\hat{A}|\psi_s\rangle = n\Delta l_p^2|\psi_s\rangle$

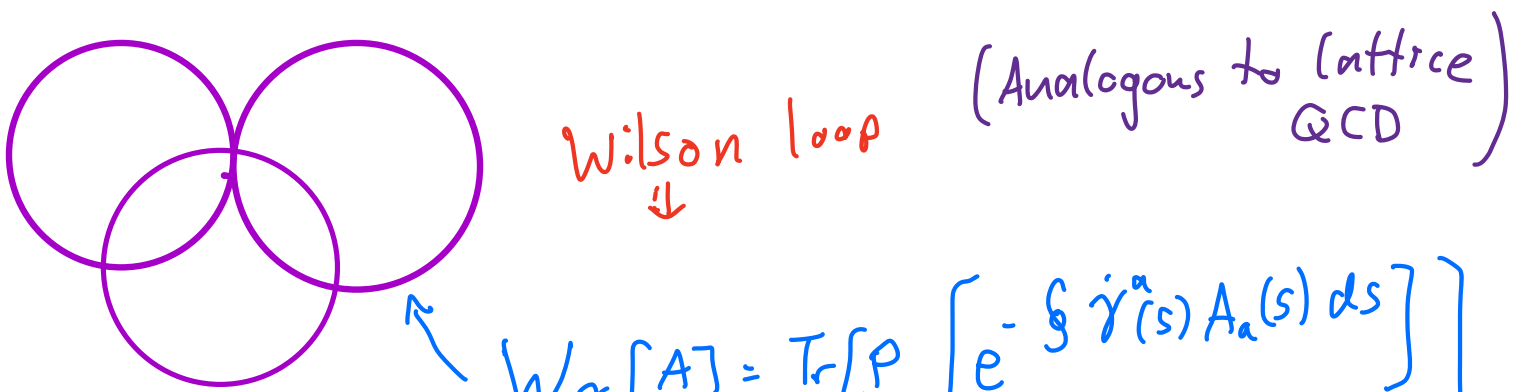


$\hat{A} = \begin{matrix} \text{purple rectangle} \\ \Delta x \end{matrix} \Delta y$   
 ↑  
 Area operator

$\hat{A}|\psi_s\rangle =$

$\approx \sqrt{3+4} \Delta l_p^2 |\psi_s\rangle$

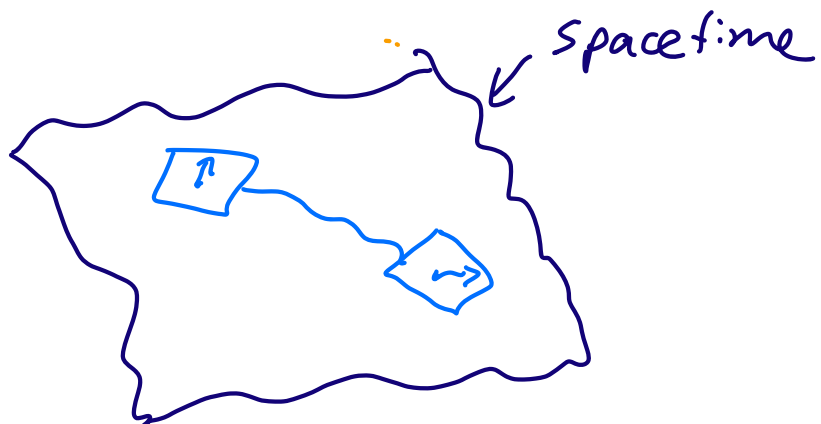
# Loop Representation



forms a basis

$$\Psi[A] = \sum_r \Psi(r) W_r[A]$$

$A_i^a$  is a Spin Connection



$$g^{ab} = E^a_i E^b_j \delta^{ij}$$

$$A_i^a = D[E_i^a]$$

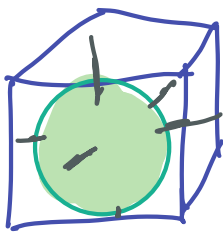
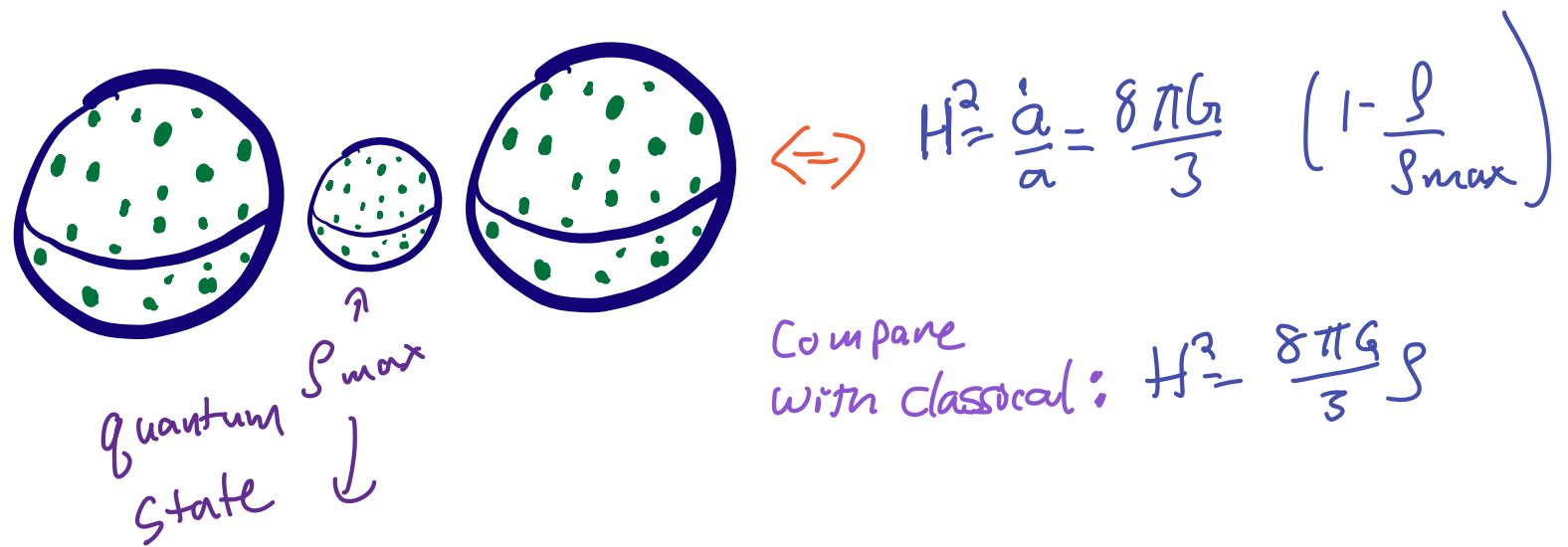
|              | Gravity                                              | EM                                                       | Weak                                                                      | Strong                                                                              |
|--------------|------------------------------------------------------|----------------------------------------------------------|---------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
|              | Metric<br>$g^{\mu\nu} = E^\mu_i E^\nu_j \delta_{ij}$ | U(1) phase<br>$e^{i\theta}  \psi\rangle$                 | SU(2) phase<br>$e^{i\theta\sigma^i}  \psi\rangle$                         | SU(3) phase<br>$e^{i\theta T^i}  \psi\rangle$                                       |
| derivative ↘ | Spacetime Connection<br>$A^\mu_i = D[E^\mu_i]$       | Photon<br>$A^\mu = (\phi, A_x, A_y, A_z)$                | W <sup>±</sup> , Z bosons<br>W <sup>±</sup> , Z                           | gluons<br>$A^i_\mu \quad i=1, \dots, 8$                                             |
| derivative ↘ | Gravitational Field<br>$R = D[A^\mu_a]$              | Electric Field<br>$F_{\mu\nu} = 2A_\mu \nu - 2\nu A_\mu$ | Weak Force<br>$G_{\mu\nu} = 2W_\mu \nu - 2\nu W_\mu + W_\mu \times W_\nu$ | Strong Force<br>$B_{\mu\nu} = 2A^i_\mu \nu - 2\nu A^i_\mu + A^i_\nu \times A^i_\mu$ |

# Loop Quantum Cosmology

Since the area eigenvalue  $\neq 0$

$$S_{\max} = \frac{1}{4\pi\gamma^2\Delta G^2\hbar} \quad S_{\max} = \frac{1}{4}$$

This excludes a big bang and implies a big bounce



Area of quantum bounded universe

$$n \Delta \ell_{Pl}^2 = 6 \Delta \ell_{Pl}^2$$

- 1.) We restrict the geometry at the bounce to be in the "ground state" to have minimum area & grav. entropy  
A.k.a Wey tensor should Vanish at initial condition
- 2.) Require perturbations to have minimum uncertainty

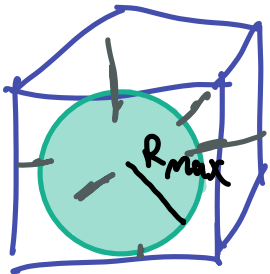
If we can ignore perturbation back reactions  
 the full quantum geometry is indistinguishable to the  
 effective dressed FLRW metric

$$\tilde{g}_{ab} = \sqrt{\frac{\langle \hat{H}_0^{-1/2} \hat{a}(t) \hat{H}_0^{-1/2} \rangle}{\langle \hat{H}_0 \rangle}} \left( \langle \hat{H}_0^{-1/2} \rangle \left( \langle \hat{H}_0^{-1/2} \hat{a}^4(t) \hat{H}_0^{-1/2} \rangle \right)^{1/2} dt + d\vec{x}^2 \right)$$

Quantum geometry becomes effectively encoded in  
 the matter field

$$\left( \nabla^2 - \frac{\partial^2}{\partial t^2} + \frac{\hat{U}}{a} \right) Q = 0 \Rightarrow \left( \nabla^2 - \frac{\partial^2}{\partial t^2} + \frac{\tilde{U}}{a} \right) Q = 0$$

with : 
$$\tilde{U}(t) = \frac{\langle \hat{H}_0^{-1/2} \hat{a}^3(t) \hat{U}(t) \hat{a}^3(t) \hat{H}_0^{-1/2} \rangle}{\langle \hat{H}_0^{-1/2} \hat{a}^4(t) \hat{H}_0^{-1/2} \rangle}$$



New length scale:

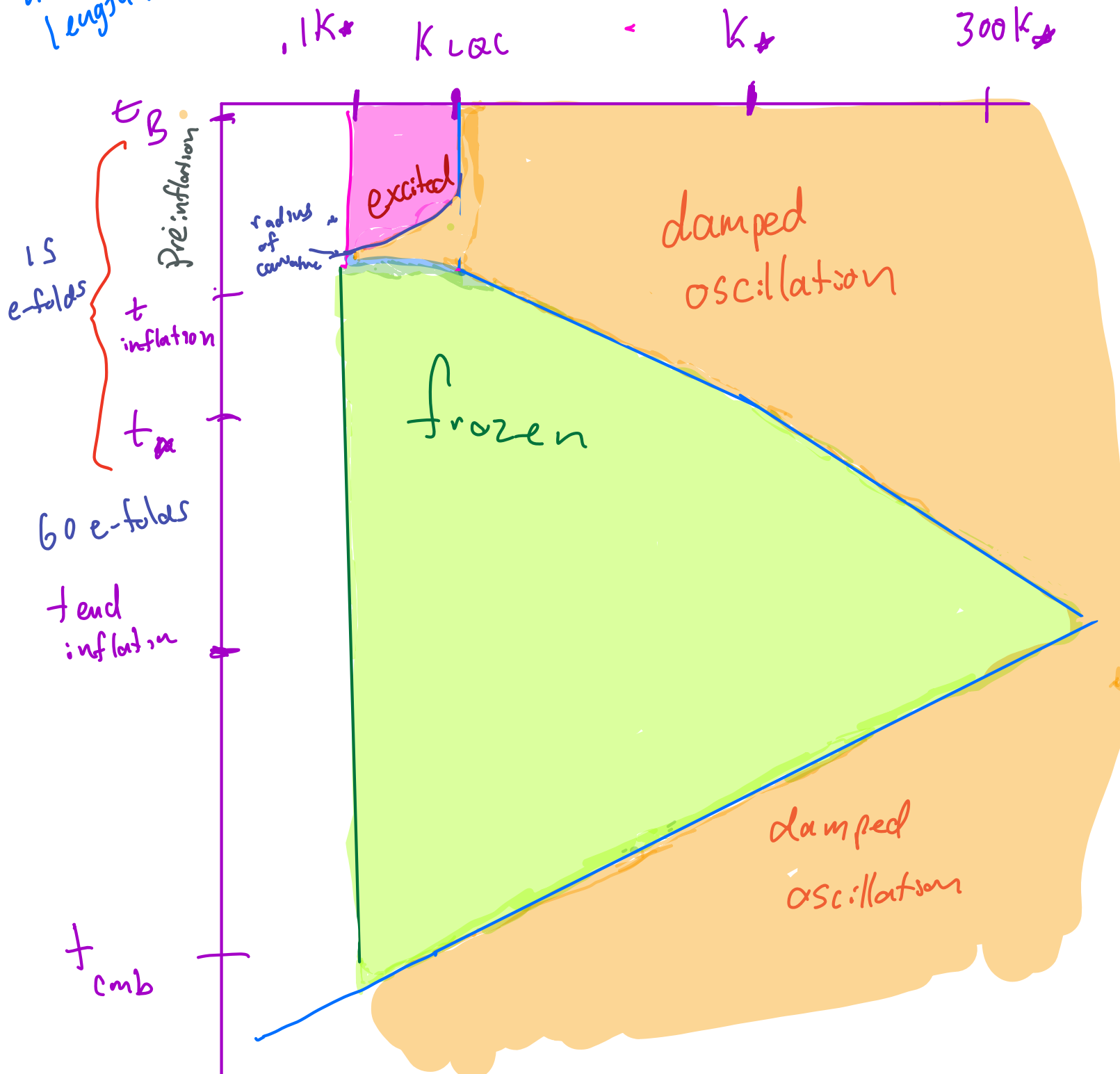
$$K_{LQC} = \left( \frac{R_{max}}{6} \right)^{1/2} = \left( \frac{62}{6 \ell_{Pl}^2} \right)^{1/2}$$

Perturbations with  $k < K_{LQC}$   
 become excited

Modified Mukhanov - Sasaki Mode:

$$\partial_\eta^2 \chi_k = a^2(\eta) \left( \frac{k^2}{\tilde{q}(\eta)^2} - \frac{\tilde{R}(\eta)}{6} \right) \chi_k$$

Hubble length line



# Observational Consequences of LQC pre-inflationary perturbations

The standard Ansatz PS:  $P_R(k) = A_s \left( \frac{k}{k_*} \right)^{n_s - 1}$

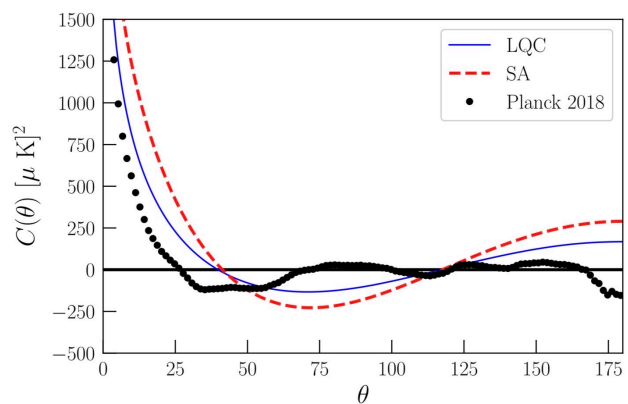
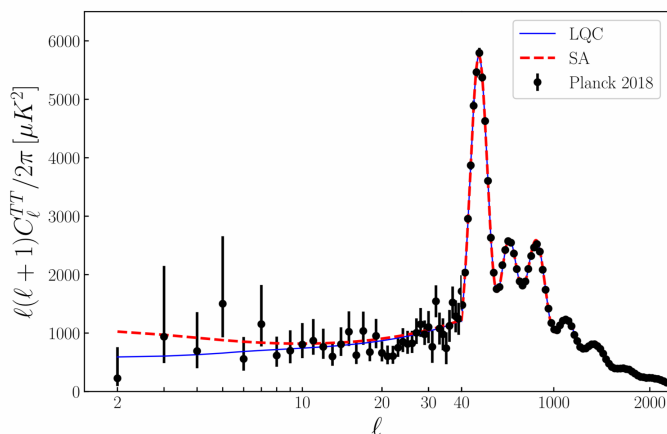
LQC Modification:  $P_R^{LQC}(k) = f(k) A_s \left( \frac{k}{k_*} \right)^{n_s - 1}$

$f(k) \approx 1$  for  $k \gg k_{LQC}$

but  $f(k) < 1$  for  $k < k_{LQC}$

$C^{TT}$  power suppression

large scale power anomaly



Ashdelfer 2020

Questions ??